

AMELIA NGUYEN

amelianguyen.ds@gmail.com | (248) 343-0064 | [linkedin.com/in/ameliang12](https://www.linkedin.com/in/ameliang12) | amelia-ng.github.io/portfolio

EDUCATION

UNIVERSITY OF MICHIGAN - DEARBORN

Dearborn, MI

Master of Science in Data Science, GPA: 4.0.

Expected Graduation: May 2027

Coursework: Machine Learning, Regression Analysis, Database Systems, Big Data, Operations Research, Data Analytics & Modeling.
Planned: Deep Learning, Advanced Artificial Intelligence, Optimization.

FOREIGN TRADE UNIVERSITY

Hanoi

Bachelor of International Business, GPA: 3.8.

May 2023

TECHNICAL SKILLS

Machine Learning & AI: Supervised Learning (Regression, Decision Trees, Random Forests, XGBoost, LightGBM, CatBoost, SVM, KNN), Unsupervised Learning (K-Means, Clustering, PCA), Deep Learning (CNN, RNN, Transformer), Reinforcement Learning, Recommendation Systems, LLMs, GenAI, Agentic AI, Hugging Face, Time Series, Computer Vision

Languages & Frameworks: Python (Pandas, Scikit-learn, NumPy, PyTorch, TensorFlow, Matplotlib, Seaborn), SQL, PySpark

Cloud & MLOps: Docker, REST APIs, Google Cloud Platform (Vertex AI, BigQuery, Cloud Storage), AWS, Spark, Hadoop, Airflow

Visualization & Analytics: Tableau, Power BI, Google Analytics, SAP Analytics Cloud

Automotive: Kalman Filtering, Robotics, Telematics, Autonomous Driving, State Estimation, Localization, Control Systems

PROFESSIONAL EXPERIENCE

UNIVERSITY OF MICHIGAN

Dearborn, MI

Graduate Teaching Assistant

Aug 2026 - Present

- Grade coursework and provide support for CIS 400 Programming Languages and CIS 541 Immersive Computing and Digital Twins.

Research Assistant

Jun 2026 - Present

- Conduct Computer Vision research on emotion recognition and physical unclonable authentication at the Trustworthy AI Lab.

GEARLAUNCH

Hanoi

Data Analyst

Nov 2023 - Jul 2025

- Served as the **sole data analyst**, managing the company's end-to-end internal analytics infrastructure and presenting business analysis insights directly to C-level leadership and department heads.
- Designed and deployed Machine Learning (Regression, Random Forest, XGBoost) and Time Series Forecasting (AR, ARIMA, SARIMAX) to support e-commerce executive strategy development, resulting in a 20% increase in annual sales KPI.
- Implemented scalable data pipelines that streamlined reporting workflows using SQL, Python, Google Cloud Platform (Cloud Run, Vertex AI), and Tableau for 35,000 monthly active user stores, resulting in an average of a 50% increase in marketing conversions.

VCCORP CORPORATION

Hanoi

Data Analyst - Team Lead

May 2023 - Nov 2023

- Managed a team of 4 to perform full-cycle A/B testing (PySpark, Python, SQL, HDFS, Elasticsearch) for the deployment of a recommendation engine across 7 digital media websites, resulting in 10–30% monthly gains in core web KPIs.
- Designed and built an automated A/B testing evaluation framework for the internal platform and a client-facing product.

Data Analyst

March 2022 - May 2023

- Provided advanced statistical analysis and Machine Learning models; designed and monitored 30+ KPIs with automated dashboards for 5 digital publishing products.
- Developed user segmentation analysis that increased user engagement with mobile app notifications by 30% within 3 months.

PROJECTS

Street Obstacle Detection | PyTorch, Computer Vision, Hugging Face

June 2026

Deployed apps for street obstacle detection using fine-tuned YOLOv11n and RT-DETR transformer-based models.

Fuel Consumption Prediction for ICE powertrain vehicles | Python, XGBoost, LightGBM, LSTM, TCN

Apr 2026 - May 2026

Achieved a 13.87% MAPE and 0.3 MAE on real-world ODB-II telematics data collected from 300 vehicles throughout a year.

Deer-Vehicle Collision Prediction | Python, Random Forest, Gradient Boosting, K-Means

Apr 2026

Predicted deer collision risk with supervised methods and K-Means clustering with PCA.

Vehicle Crash Frequency and Severity Prediction | Python, Regression, Classification

Nov 2025 - Dec 2025

Deployed predictive modeling and diagnostics using linear regression and logistic regression with 85% accuracy.

CERTIFICATIONS

Self-Driving Cars Specialization (University of Toronto Online) | Simulation, Control System, AI

Jul 2026 - Present

AI for Autonomous Vehicles and Robotics (University of Michigan Online) | Robotics, Autonomous Driving Systems

Jul 2026